

Project 1 - Random Number Generator

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Develop a random number generator using a quantum circuit. Consider generating a positive integer from 0 to 1000. After measuring the quantum circuit, take the result and convert it into an integer. Unlike classical computing, which uses pseudorandom number generation, quantum computing enables the generation of truly random numbers. For security-critical applications, using a quantum device to generate numbers can be a significant advantage.

Consider two scenarios:

1. **Uniform:** the number is generated according to a uniform distribution (all values have the same probability).
2. **Biased:** the distribution favors higher values. Hint: use S , T , or R_z gates to obtain the desired effect.

Instructions

1. The quantum circuit must be responsible for generating random bit strings, which can then be converted into integers.
2. Implement two quantum circuits, one for each scenario.
3. In scenario (2), the bias should be defined by the group.
4. Execute the quantum circuit several times and verify the distributions for both scenarios. Execute in the simulator and the IBM computer. Compare the results.